PROGRAM NO: 1

Aim:Write a program to design a login form with username and password using linear layout and toast valid credentials.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textPersonName"
        android:hint="username"
        android:textColor="#E91E63"
        android:layout_margin="20dp"
        android:id="@+id/username"
        />
```

```
    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textPassword"
        android:hint="password"
        android:textColor="#E91E63"
        android:layout_margin="20dp"
        android:id="@+id/password"
        />
<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/login"
    android:text="login"
    android:layout_margin="10dp"
    />
</LinearLayout>
```

MainActivity.java

```
package com.example.loginapp;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;
```

```
import android.widget.EditText;

import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
EditText username;
EditText password;
Button login;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    //variable initialization

    username=findViewById(R.id.username);
    password=findViewById(R.id.password);
    login=findViewById(R.id.login);
    login.setOnClickListener(new View.OnClickListener() {
      @Override
      public void onClick(View view) {


        if (username.getText().toString().equals("admin")&&
password.getText().toString().equals("admin")){
          Toast.makeText(MainActivity.this,"Login
Successful",Toast.LENGTH_SHORT).show();
        }
      }
```

```
    });
  }
}
```

Aim:Write a program that demonstrates Activity Lifecycle.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Question Explaining the Activity Lifecycle"
        />
</LinearLayout>
```

MainActivity.java

```
package com.example.activitylifecycle;

import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
import android.util.Log;
import android.widget.Toast;


public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Log.d("lifecycle","onCreate invoked");
        Toast.makeText(getApplicationContext(),"onCreate
invoked",Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onStart()
    {
        super.onStart();
        Log.d("lifecycle","onStart invoked");
        Toast.makeText(getApplicationContext(),"onStart
invoked",Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onResume()
    {
        super.onResume();
        Log.d("lifecycle","onResume invoked");
```

```
    Toast.makeText(getApplicationContext(),"onResume
invoked",Toast.LENGTH_SHORT).show();
  }

  @Override
  protected void onPause()
  {
    super.onPause();
    Log.d("lifecycle","onPause invoked");
    Toast.makeText(getApplicationContext(),"onPause
invoked",Toast.LENGTH_SHORT).show();
  }

  @Override
  protected void onStop()
  {
    super.onStop();
    Log.d("lifecycle","onStop invoked");
    Toast.makeText(getApplicationContext(),"onStop
invoked",Toast.LENGTH_SHORT).show();
  }

@Override
  protected void onRestart()
  {
    super.onRestart();
    Log.d("lifecycle","onRestart invoked");
    Toast.makeText(getApplicationContext(),"onRestart
```

```
    invoked",Toast.LENGTH_SHORT).show();

  }

  @Override
  protected void onDestroy()
  {
    super.onDestroy();
    Log.d("lifecycle","onDestroy invoked");
    Toast.makeText(getApplicationContext(),"onDestroy
invoked",Toast.LENGTH_SHORT).show();

  }

}
```

Aim:Write a program implementing basic arithmetic operations of a simple calculator.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/number1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number 01"
        android:inputType="numberDecimal"/>

    <EditText
        android:id="@+id/number2"
        android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:hint="Enter Number 02"
    android:inputType="numberDecimal"/>


<TextView
    android:id="@+id/result_text"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:textColor="@color/black"
    android:textSize="17sp"
    android:textStyle="bold"/>

<Button
    android:id="@+id/add_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="+"/>

<Button
    android:id="@+id/sub_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="-"/>

<Button
```

```
    android:id="@+id/mul_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="x"/>
  <Button
    android:id="@+id/div_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="/"/>
  <Button
    android:id="@+id/clear_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="clear"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.calculator;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
```

```
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {


    EditText number1, number2;
    TextView result_text;
    Button add_btn, sub_btn, mul_btn, div_btn, clear_btn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        number1 = findViewById(R.id.number1);
        number2 = findViewById(R.id.number2);
        result_text = findViewById(R.id.result_text);
        add_btn = findViewById(R.id.add_btn);
        sub_btn = findViewById(R.id.sub_btn);
        mul_btn = findViewById(R.id.mul_btn);
        div_btn = findViewById(R.id.div_btn);
        clear_btn = findViewById(R.id.clear_btn);

        add_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String number1_text = number1.getText().toString();
                String number2_text = number2.getText().toString();
                int num1 = Integer.parseInt(number1_text);
                int num2 = Integer.parseInt(number2_text);
```

```
            float sum
                    = num1 + num2;
            result_text.setText("The addition of the two numbers is:" + sum);
            Toast.makeText(getApplicationContext(), "The addition of the two numbers is:" + sum,
Toast.LENGTH_SHORT).show();
        }
    });

    sub_btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            String number1_text = number1.getText().toString();
            String number2_text = number2.getText().toString();
            int num1 = Integer.parseInt(number1_text);
            int num2 = Integer.parseInt(number2_text);
            float sub = num1 - num2;
            result_text.setText("The substraction of the two numbers is:" + sub);
            Toast.makeText(getApplicationContext(), "The substraction of the two numbers is:" +
sub, Toast.LENGTH_SHORT).show();
        }
    });

    mul_btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            String number1_text = number1.getText().toString();
            String number2_text = number2.getText().toString();
            int num1 = Integer.parseInt(number1_text);
            int num2 = Integer.parseInt(number2_text);
            float mul = num1 * num2;
```

```
            result_text.setText("The multiplication of the two numbers is:" + mul);


            Toast.makeText(getApplicationContext(), "The multiplication of the two numbers is:" +
mul, Toast.LENGTH_SHORT).show();
        }



    });

    div_btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            String number1_text = number1.getText().toString();
            String number2_text = number2.getText().toString();
            int num1 = Integer.parseInt(number1_text);
            int num2 = Integer.parseInt(number2_text);
            float div = num1 / num2;
            result_text.setText("The division of the two numbers is:" + div);
            Toast.makeText(getApplicationContext(), "The division of the two numbers is:" + div,
Toast.LENGTH_SHORT).show();
        }

    });

    clear_btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            number1.setText("");
            number2.setText("");
            result_text.setText(("));
```

```
            Toast.makeText(getApplicationContext(), "Inputs cleared...",
Toast.LENGTH_SHORT).show();

        }
    });
  }

}
```

Aim:Write a program that implements validations on various UI controls.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_gravity="center_horizontal"
    android:orientation="vertical"
    android:paddingHorizontal="20dp"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="User Form"
        android:textSize="20sp"
        android:layout_marginTop="30dp"
        android:textColor="@color/black"
        android:textStyle="bold"/>
    <EditText
        android:id="@+id/name_et"
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:hint="Full Name"
    android:inputType="textPersonName"
    android:minHeight="48dp"
    android:textColorHint="#757575"
    android:importantForAutofill="no"/>
<EditText
    android:id="@+id/email_et"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:hint="Email ID"
    android:inputType="textEmailAddress"
    android:minHeight="48dp"
    android:textColorHint="#757575"
    android:importantForAutofill="no"/>
<EditText
    android:id="@+id/age_et"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:hint="Age"
    android:inputType="number"
    android:minHeight="48dp"
    android:textColorHint="#757575"
    android:importantForAutofill="no"/>
<Button
    android:id="@+id/submit_btn"
```

```
        android:layout_width="268dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Submit"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.myui;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    EditText name_et,email_et,age_et;
    Button submit_btn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
setContentView(R.layout.activity_main);
name_et=findViewById(R.id.name_et);
email_et=findViewById(R.id.email_et);
age_et=findViewById(R.id.age_et);
submit_btn=findViewById(R.id.submit_btn);
submit_btn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        name_et.setError(null);
        age_et.setError(null);
        email_et.setError(null);
        String name_txt=name_et.getText().toString();
        String email_txt=email_et.getText().toString();
        String age_txt=age_et.getText().toString();

        if(name_txt.equals("")) {
            name_et.setError("Please enter your name");
            name_et.requestFocus();
        } else if (email_txt.equals("")) {
            email_et.setError("Please enter your Email ID");
            email_et.requestFocus();

        }
        else if (age_txt.equals("")) {
            age_et.setError("Please enter your age");
            age_et.requestFocus();

        }
        else {
            Toast.makeText(getApplicationContext(),"Form Submitted
```

```
Successfully",Toast.LENGTH_SHORT).show();

                }
            }
        });
    }
}
```

Aim:Write a program to design a registration activity and store registration details in local memory of phone using Intents and SharedPreferences.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:paddingHorizontal="20dp"
    android:id="@+id/main_layout"
    android:paddingVertical="10dp"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Registration Form"
        android:textStyle="bold"
        android:textColor="@color/black"
        android:textSize="20sp" />
```

```
<EditText
    android:id="@+id/fullname"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:inputType="textPersonName"
    android:layout_marginTop="30dp"
    android:hint="Full Name"/>

<EditText
    android:id="@+id/emailid"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:inputType="textEmailAddress"
    android:layout_marginTop="10dp"
    android:hint="Email ID"/>

<EditText
    android:id="@+id/password"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:inputType="textPassword"
    android:hint="Password"/>

<Button
    android:id="@+id/register_btn"
    android:layout_width="201dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="Register"/>
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.intends;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.util.Patterns;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.LinearLayout;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText fullname,emailid,password;
    Button register_btn;
    LinearLayout main_layout;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
fullname=findViewById(R.id.fullname);
emailid=findViewById(R.id.emailid);
password=findViewById(R.id.password);
register_btn=findViewById(R.id.register_btn);
main_layout=findViewById(R.id.main_layout);

register_btn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {

        fullname.setError(null);
        emailid.setError(null);
        password.setError(null);

        String password_regex = "^(?=.*[0-9])(?=.*[a-z])(?=.*[A-
         Z])(?=.*[@#$%^&+=])(?=\\S+$).{4,}$";
        String fullname_text = fullname.getText().toString();
        String emailid_text = emailid.getText().toString();
        String password_text = password.getText().toString();

        if (fullname_text.equals("")) {
            fullname.requestFocus();
            fullname.setError("Please enter fullname!!");
        } else if (emailid_text.equals("")) {
            emailid.requestFocus();
            emailid.setError("Please enter emaild-id!!");

        } else if (!Patterns.EMAIL_ADDRESS.matcher(emailid_text).matches()) {
            emailid.requestFocus();
```

```
            emailid.setError("Please enter a valid email-id!!");

        } else if (!password_text.matches(password_regex)) {
            password.requestFocus();
            password.setError("Password should contain -\n a digit must occur at least once \n a lower case letter must occur at least once \n an upper case letter occur at least once \n a special character like @#$%^&+=\n No blank spaces allowed \n at least 6 characters");

        } else {
            SharedPreferences pref = getSharedPreferences("register_data", MODE_PRIVATE);
            SharedPreferences.Editor pref_edit = pref.edit();
            pref_edit.putString("reg_fullname", fullname_text);
            pref_edit.putString("reg_emailid", emailid_text);
            pref_edit.putString("reg_password", password_text);
            pref_edit.apply();

            Intent intent = new Intent(getApplicationContext(),IntendS2.class);
            startActivity(intent);
        }
    }
});

}
}
```

activity_intend_s2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="10dp"
    tools:context=".IntendS2">

    <TextView
        android:id="@+id/fullname_result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>

    <TextView
        android:id="@+id/emailid_result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>

    <TextView
        android:id="@+id/password_result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

IntendS2.java

```
package com.example.intends;

import androidx.appcompat.app.AppCompatActivity;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.TextView;

public class IntendS2 extends AppCompatActivity {

    TextView fullname_result,emailid_result,password_result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_intend_s2);

        fullname_result=findViewById(R.id.fullname_result);
        emailid_result=findViewById(R.id.emailid_result);
        password_result=findViewById(R.id.password_result);

        SharedPreferences pref=getSharedPreferences("register_data",MODE_PRIVATE);
        String name=pref.getString("reg_fullname","Not Available!!");
        String email=pref.getString("reg_emailid","Not Available!!");
        String password=pref.getString("reg_password","Not Available!!");
        fullname_result.setText(name);
        emailid_result.setText(email);
        password_result.setText(password);
    }
}
```

Aim:Write a program to design a Simple Calculator using GridLayout and Cascade LinearLayout.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:id="@+id/calculator_mainlay"
    android:padding="20dp"
    tools:context=".MainActivity">
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="30dp"
    android:hint="number 01"
    android:inputType="number"/>
```

```
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:hint="number 02"
    android:inputType="number"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:hint="Result"
    android:clickable="false"
    android:enabled="false"
    android:inputType="number"/>
<LinearLayout
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:orientation="vertical"
     android:layout_marginTop="20dp"
     android:gravity="bottom">
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
```

```
    android:weightSum="4">
<Button
    android:id="@+id/numclearall_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="CE"
    android:layout_weight="1"/>
<Button
   android:id="@+id/numclear_btn"
   android:layout_width="0dp"
   android:layout_height="60dp"
   android:textColor="@color/white"
   android:text="C"
   android:layout_marginStart="10dp"
   android:layout_weight="1"/>
<Button
   android:id="@+id/nummodulus_btn"
   android:layout_width="0dp"
   android:layout_height="60dp"
   android:textColor="@color/white"
   android:text="%"
   android:layout_marginStart="10dp"
   android:layout_weight="1"/>
```

```
<Button
  android:id="@+id/divide_btn"
  android:layout_width="0dp"
  android:layout_height="60dp"
  android:textColor="@color/white"
  android:text="/"
  android:layout_marginStart="10dp"
  android:layout_weight="1"/>

 </LinearLayout>

 <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginTop="10dp"
    android:weightSum="4">
  <Button
    android:id="@+id/num7_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="7"
    android:layout_weight="1"/>
```

```
<Button
    android:id="@+id/num8_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="8"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>

<Button
    android:id="@+id/num9_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="9"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>

<Button
    android:id="@+id/multiply_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
```

```
    android:text="*"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
</LinearLayout>
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginTop="10dp"
    android:weightSum="4">
<Button
    android:id="@+id/num4_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="4"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
<Button
    android:id="@+id/num5_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="5"
```

```
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
<Button
    android:id="@+id/num6_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="6"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
<Button
    android:id="@+id/subtract_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="-"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
</LinearLayout>
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginTop="10dp"
```

```
    android:weightSum="4">
<Button
    android:id="@+id/num1_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="1"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
<Button
    android:id="@+id/num2_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="2"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
<Button
    android:id="@+id/num3_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="3"
    android:layout_marginStart="10dp"
```

```
    android:layout_weight="1"/>
<Button
    android:id="@+id/add_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="+"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
</LinearLayout>
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginTop="10dp"
    android:weightSum="4">
<Button
    android:id="@+id/numplusminus_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="+/-"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
```

```
<Button
    android:id="@+id/num0_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="0"
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
<Button
    android:id="@+id/decimal_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="."
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
<Button
    android:id="@+id/equals_btn"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:textColor="@color/white"
    android:text="="
    android:layout_marginStart="10dp"
    android:layout_weight="1"/>
```

```
  </LinearLayout>
   </LinearLayout>
</LinearLayout>
```

MainActivity.java

```
package com.example.calcapplication;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
   }
}
```

activity_ques06_grid_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/calculator_mainlay"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".Ques06GridLayoutActivity">

<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="30dp"
    android:hint="Number 01"
    android:inputType="number"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:hint="Number 02"
    android:inputType="number"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:clickable="false"
    android:enabled="false"
    android:hint="Result"
    android:inputType="number"/>
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
    android:gravity="bottom"
    android:orientation="vertical">
<GridLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:columnCount="4"
    android:orientation="horizontal"
    android:rowCount="5">

<Button
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="CE"
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="C"
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="%"
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="/"
```

```
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="7"
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="8"
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="9"
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="*"
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="4"
    android:textColor="@color/white"/>
<Button
```

```
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="5"
    android:textColor="@color/white"/>
<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="6"
    android:textColor="@color/white"/>

<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="-"
    android:textColor="@color/white"/>

<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="1"
    android:textColor="@color/white"/>

<Button
    android:layout_marginStart="10dp"
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="2"
    android:textColor="@color/white"/>
```

```
<Button
  android:layout_marginStart="10dp"
  android:layout_marginTop="20dp"
  android:padding="13dp"
  android:text="3"
  android:textColor="@color/white"/>
<Button
  android:layout_marginStart="10dp"
  android:layout_marginTop="20dp"
  android:padding="13dp"
  android:text="+"
  android:textColor="@color/white"/>
<Button
  android:layout_marginStart="10dp"
  android:layout_marginTop="20dp"
  android:padding="13dp"
  android:text="+/-"
  android:textColor="@color/white"/>
<Button
  android:layout_marginStart="10dp"
  android:layout_marginTop="20dp"
  android:padding="13dp"
  android:text="0"
  android:textColor="@color/white"/>
<Button
  android:layout_marginStart="10dp"
  android:layout_marginTop="20dp"
  android:padding="13dp"
  android:text="."
  android:textColor="@color/white"/>
<Button
  android:layout_marginStart="10dp"
```

```
    android:layout_marginTop="20dp"
    android:padding="13dp"
    android:text="="
    android:textColor="@color/white"/>
</GridLayout>

</LinearLayout>



</LinearLayout>
```

Ques06GridLayoutActivity.java

```
package com.example.calcapplication;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class Ques06GridLayoutActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_ques06_grid_layout);
    }
}
```

Aim:Write a program to create a Facebook page using RelativeLayout,set properties using .xml file.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:orientation="vertical"
    android:layout_height="match_parent"
    android:id="@+id/mainlay"
    tools:context=".MainActivity">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:src="@drawable/facebook"
        android:scaleType="centerCrop"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Facebook"
```

```
    android:textAlignment="center"
    android:layout_marginTop="5dp"
    android:layout_gravity="center"
    />

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:paddingHorizontal="30dp"
    android:layout_marginTop="20dp"
    android:orientation="vertical"
    />


<EditText
    android:id="@+id/fb_id"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Phone or email"
    android:digits=""
    android:inputType="text"/>


<EditText
    android:id="@+id/fb_pass"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:hint="Password"
    android:inputType="textPassword"/>
```

```
<Button
    android:id="@+id/fb_loginbtn"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:text="Log In"
    android:layout_marginTop="20dp"
    android:textAllCaps="false"
    android:backgroundTint="#3F51B5"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Forgot Password?"
    android:textStyle="bold"
    android:textColor="#3F51B5"
    android:textSize="17sp"
    android:layout_marginTop="10dp"/>

<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Create New"
    android:layout_marginTop="20dp"
    android:textAllCaps="false"
    android:backgroundTint="#4CAF50"/>

</LinearLayout
```

MainActivity.java

```
package com.example.applications;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity{

@Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
  }
}
```

Aim:Write a program to develop an application that toggles image using FrameLayout.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <ToggleButton
        android:id="@+id/toggle_img_btn"
        android:layout_width="60dp"
        android:layout_height="60dp"
        android:textOff=""
        android:textOn=""
        android:background="@drawable/baseline_volume_up_24"
        android:layout_gravity="center"/>

</FrameLayout>
```

MainActivity.java

```
package com.example.toggle;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.widget.Toast;
import android.widget.ToggleButton;

public class MainActivity extends AppCompatActivity {
   ToggleButton toggle_img_btn;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      toggle_img_btn=findViewById(R.id.toggle_img_btn);

      toggle_img_btn.setOnClickListener(v->{
         if(toggle_img_btn.isChecked()){

toggle_img_btn.setBackgroundDrawable(getResources().getDrawable(R.drawable.baseline_volum
e_up_24));
            Toast.makeText(getApplicationContext(),"Volume
Up.",Toast.LENGTH_SHORT).show();

         }
         else{
```

```
toggle_img_btn.setBackgroundDrawable(getResources().getDrawable(R.drawable.baseline_volum
e_mute_24));

            Toast.makeText(getApplicationContext(),"Volume
Mute.",Toast.LENGTH_SHORT).show();

        }

    });

  }

}
```

Aim:Write a program to implement Adapters and perform Exception handling.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Exception Activity"
        />
    <TextView
        android:id="@+id/textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
```

```
    android:text="Value of 0/0"
    />

</LinearLayout>
```

MainActivity.java

```
package com.example.adapter;

import androidx.appcompat.app.AppCompatActivity;

import  android.os.Bundle;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
  TextView textView;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    textView=findViewById(R.id.textview);

    try {
      int n1=0,n2=0;
      int a=n1/n2;
      textView.setText("Value of 0/0:"+a);
```

```
        Toast.makeText(getApplicationContext(),"The value
is:"+a,Toast.LENGTH_SHORT).show();




    }
    catch(Exception e){
        Toast.makeText(getApplicationContext(),"The caught exception
is:"+e.getMessage(),Toast.LENGTH_LONG).show();
    }
  }
}
```

Aim:Write a program to implement Intent to navigate between multiple activities.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Home Page"
        android:textSize="17sp" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="View Profile"
```

```
        android:layout_marginTop="1dp"/>

    </LinearLayout>
```

MainActivity.java

```
package com.example.navigate;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {


    Button button;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        button=findViewById(R.id.button);
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
```

```
            Intent intent=new Intent(getApplicationContext(),Navigate2.class);
            startActivity(intent);

        }
      });
    }
  }
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".Navigate2">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:src="@drawable/baseline_person_pin_24"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Person Profile"
        android:textColor="@color/black"
```

```
        android:textStyle="bold"
        android:textSize="17sp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.navigate;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class Navigate2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_navigate2);
    }
}
```

Aim:Write a program to develop application that works with explicit intents.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome" />

    <Button
        android:id="@+id/goto_second_btn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="admin"
        android:layout_marginTop="10dp"/>
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.explicit;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    Button goto_second_btn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        goto_second_btn=findViewById(R.id.goto_second_btn);
        goto_second_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent=new Intent(getApplicationContext(),Explicit2.class);
                startActivity(intent);
            }
        });
    }}
```

SecondActivity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".Explicit2">


    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to trip"/>

    <Button
        android:id="@+id/goto_third_btn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="move to page"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

SecondActivity.java

```
package com.example.explicit;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class Explicit2 extends AppCompatActivity {


    Button goto_third_btn;



    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_explicit2);


        goto_third_btn=findViewById(R.id.goto_third_btn);
        goto_third_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent=new Intent(getApplicationContext(),Explicit3.class);
                startActivity(intent);
            }
        });
```

```
}
}
```

ThirdActivity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".Explicit3">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to trip"/>

    <Button
        android:id="@+id/goto_main_btn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go back"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

ThirdActivity.java

```
package com.example.explicit;
```

```
import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class Explicit3 extends AppCompatActivity {

    Button goto_main_btn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_explicit3);

        goto_main_btn=findViewById(R.id.goto_main_btn);
        goto_main_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent=new Intent(getApplicationContext(),MainActivity.class);
                startActivity(intent);
            }
        });
    }
}
```

Aim:Write a program to implement Option Menu to navigate to activities.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools" android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="MenuOptions Menu Program"
        />
</LinearLayout>
```

MainActivity.java

```
package com.example.menuoption;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int menu_id=item.getItemId();
        switch (menu_id){
            default: {
                break;
            }
        }
        return true;
    }
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
```

```
        getMenuInflater().inflate(R.menu.menu_item,menu);
        return super.onCreateOptionsMenu(menu);
    }
}
```

Menu_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
<item
   android:id="@+id/menu_option_1"
   android:title="Page1" />
<item
   android:id="@+id/menu_option_2"
   android:title="Page2" />
</menu>
```

PROGRAM NO: 13

Aim:Write a program to develop an application that uses ArrayAdapter with ListView.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ListView
        android:id="@+id/listview"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.arrayadapter;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        ListView listview;
        String[]person_qualify={"Item1", "Item2", "Item3", "Item4", "Item5", "Item6", "Item7",
"Item8", "Item9", "Item10", "Item11", "Item12"};
        listview=findViewById(R.id.listview);
        listview.setAdapter(new ArrayAdapter(getApplicationContext(),
                android.R.layout.simple_expandable_list_item_1,person_qualify));
        listview.setOnItemClickListener((parent,view,position,id)-> {
            Toast.makeText(this,"The selected item is-" +person_qualify[position],
```

```
                Toast.LENGTH_SHORT).show();
        });
    }
}
```

PROGRAM NO :14

Aim:Write a program to develop an application that uses GridView with images and display Alert box on selection.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="My Apps"
        android:layout_marginTop="30dp"
        android:textColor="@color/black"
        android:textSize="20sp"
        android:textStyle="bold"/>

    <GridLayout
        android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
android:layout_marginTop="30dp"
android:columnCount="3"
android:orientation="horizontal"
android:rowCount="3">

<ImageView
    android:id="@+id/wifi_btn"
    android:layout_width="80dp"
    android:layout_height="80dp"
    android:padding="13dp"
    android:src="@drawable/baseline_wifi_24"
    android:textColor="@color/white"/>
<ImageView
    android:id="@+id/bluetooth_btn"
    android:layout_width="80dp"
    android:layout_height="80dp"
    android:layout_marginStart="10dp"
    android:padding="13dp"
    android:src="@drawable/baseline_bluetooth_24"
    android:textColor="@color/white"/>
<ImageView
    android:id="@+id/volume_btn"
    android:layout_width="80dp"
    android:layout_height="80dp"
    android:layout_marginStart="10dp"
    android:padding="13dp"
    android:src="@drawable/baseline_volume_up_24"
    android:textColor="@color/white"/>
```

```
    </GridLayout>
</LinearLayout>
```

MainActivity.java

```
package com.example.gridview;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

import android.content.DialogInterface;
import android.os.Bundle;
import android.view.View;
import android.widget.ImageButton;
import android.widget.ImageView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    ImageView wifi_btn,bluetooth_btn,volume_btn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        wifi_btn=findViewById(R.id.wifi_btn);
        bluetooth_btn=findViewById(R.id.bluetooth_btn);
        volume_btn=findViewById(R.id.volume_btn);
```

```
    wifi_btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            myAlertBox("Album Option");
        }
    });

    bluetooth_btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            myAlertBox("Mail Option");
        }
    });

    volume_btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            myAlertBox("Money Option");
        }
    });

}

public void myAlertBox(String buttonname) {
    AlertDialog.Builder builder=new AlertDialog.Builder(MainActivity.this);
    builder.setMessage(("Click OK to select option"));
    builder.setTitle("Alert!");
    builder.setCancelable(false);
    builder.setPositiveButton("Yes", new DialogInterface.OnClickListener() {
```

```
        @Override
        public void onClick(DialogInterface dialogInterface, int i) {
            Toast.makeText(getApplicationContext(),buttonname+"is successfully
selected",Toast.LENGTH_SHORT).show();

        }
    });
```

```
    builder.setNegativeButton("No",(DialogInterface.OnClickListener)(dialog, which)->{
        dialog.cancel();
    });

    AlertDialog alertDialog= builder.create();
    alertDialog.show();
  }
}
```

Aim:Write a program to develop an application that implements Spinners components and perform event handling.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Birds Spinner"
        android:textStyle="bold"
        android:textSize="20sp"
        android:textColor="@color/black"/>
    <Spinner
        android:id="@+id/spinner"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:layout_margin="10dp"/>

    </LinearLayout>
```

MainActivity.java

```
    package com.example.spinner;

    import androidx.appcompat.app.AppCompatActivity;


    import android.os.Bundle;
    import android.view.View;
    import android.widget.AdapterView;
    import android.widget.ArrayAdapter;
    import android.widget.Spinner;
    import android.widget.Toast;

    public class MainActivity extends AppCompatActivity {
       Spinner spinner;
       String[] designations={"Select a bird", "Parrot", "Crow", "Pigeon", "Eagle", "Cuckoo"};


       @Override
       protected void onCreate(Bundle savedInstanceState) {
          super.onCreate(savedInstanceState);
          setContentView(R.layout.activity_main);

          spinner=findViewById(R.id.spinner);
          spinner.setSelection(0,false);
```

```
    ArrayAdapter adapter=new ArrayAdapter(this, android.R.layout.simple_spinner_item,
designations);
    adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
    spinner.setAdapter(adapter);

    spinner.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
        @Override
        public void onItemSelected(AdapterView<?> adapterView, View view, int i, long l) {
          Toast.makeText(getApplicationContext(),"Selected bird is:"+designations[i],
Toast.LENGTH_SHORT).show();
        }

        @Override
        public void onNothingSelected(AdapterView<?> adapterView) {

        }
    });
  }
}
```

Program 16

Write a program to create database using SQLite and perform INSERT and SELECT.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Database Operations"
        android:layout_gravity="center"
        android:layout_marginTop="50dp"
        android:textSize="25sp"
        android:textStyle="bold"/>
    <EditText
```

```
        android:id="@+id/rollno"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Roll no"
        android:layout_marginHorizontal="20dp"

       android:layout_marginTop="30dp"/>
    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Name"
        android:layout_marginHorizontal="20dp"
        android:layout_marginTop="10dp"/>
    <EditText
        android:id="@+id/email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Email ID"
        android:layout_marginHorizontal="20dp"
        android:layout_marginTop="10dp"/>

<Button
        android:id="@+id/insert_btn"
        android:layout_width="286dp"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:layout_marginTop="30dp"
        android:backgroundTint="#8BC34A"
        android:text="Insert into table" />
```

```
    <Button
        android:id="@+id/select_btn"
        android:layout_width="286dp"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:layout_marginTop="30dp"
        android:backgroundTint="#8BC34A"
        android:text="View from table"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.sqlite;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity{
    EditText rollno, name, email;
    Button insert_btn, select_btn;
    dhelpher db;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        rollno=findViewById(R.id.rollno);
        name=findViewById(R.id.name);
        email=findViewById(R.id.email);
        insert_btn=findViewById(R.id.insert_btn);
        select_btn = findViewById(R.id.select_btn);
        db=new dhelpher(getApplicationContext());

        insert_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                int rollno_num = Integer.parseInt(rollno.getText().toString());
                String name_txt = name.getText().toString();
                String email_txt = email.getText().toString();



                boolean insert_result = db.insertToDB(rollno_num, name_txt,
email_txt);
                if (insert_result) {
                    Toast.makeText(getApplicationContext(), "Inserted
successfully."
                            Toast.LENGTH_LONG).show();
                }
                else {
                    Toast.makeText(getApplicationContext(), "Insertion
failed!!",Toast.LENGTH_LONG).show();
```

```
            }

        }
    });

    select_btn.setOnClickListener(new View.OnClickListener() {

        @Override
        public void onClick(View view) {
            Cursor res = db.selectFromDB();
            if (res.getCount() == 0) {
                Toast.makeText(getApplicationContext(), "No entry Exist",
Toast.LENGTH_LONG).show();
            } else {
                StringBuffer buffer = new StringBuffer();
                while (res.moveToNext()){
                    buffer.append("rollno:" + res.getString(0) + "\n");
                    buffer.append("Name:" + res.getString(1) + "\n");
                    buffer.append("email:" + res.getString(2) + "\n");
                }
                AlertDialog.Builder builder = new
AlertDialog.Builder(MainActivity.this);
                builder.setCancelable(true);
                builder.setTitle("User Entries");



                builder.setMessage(buffer.toString());
                builder.show();
```

```
            }
        }
    });
  }
}
```

DBhelpher.java

```
package com.example.sqlite;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import androidx.annotation.Nullable;
public class dhelpher extends SQLiteOpenHelper {
  public dhelpher(@Nullable Context context) {
    super(context, "MyDB", null, 1);
  }

  @Override
  public void onCreate(SQLiteDatabase sqliteDatabase) {
    sqliteDatabase.execSQL("CREATE TABLE userdetails (rollno
INTEGER PRIMARY KEY, name TEXT, email TEXT)");
  }
```

```
@Override
public void onUpgrade(SQLiteDatabase sqLiteDatabase,int i,int i1){
   sqLiteDatabase.execSQL("Drop table if exists userdetails");
}
public boolean insertToDB(int rollno, String name, String email) {
   SQLiteDatabase db = this.getWritableDatabase();
   ContentValues values = new ContentValues();
   values.put("rollno", rollno);
   values.put("name", name);
   values.put("email", email);
   long result = db.insert("userdetails", null, values);
   if (result >= 0) {
      return true;
   } else {
      return false;
   }
}

public Cursor selectFromDB() {
   SQLiteDatabase DB = this.getWritableDatabase();
   Cursor cursor = DB.rawQuery("Select * from userdetails", null);
   return cursor;
}

public boolean updateToDB(int rollno, String name, String email) {
   SQLiteDatabase db = this.getWritableDatabase();
   ContentValues values = new ContentValues();
   values.put("name", name);
   values.put("email", email);
```

```
        Cursor check_user = db.rawQuery("SELECT * from userdetails
WHERE rollno=?", new
                String[]{String.valueOf(rollno)});
        if (check_user.getCount() > 0) {



            long update_user_query = db.update("userdetails", values,
"rollno=?", new String[]{String.valueOf(rollno)});

            if (update_user_query >= 0) {
                return true;
            } else {
                return false;
            }
        } else{

            return false;

        }
    }

    public boolean deleteFromDB(int rollno)
    {
        SQLiteDatabase db=this.getWritableDatabase();
        Cursor check_user=db.rawQuery("SELECT * FROM userdetails
WHERE rollno=?",new
                String[] {String.valueOf(rollno)});
        if(check_user.getCount()>(0)){
            long delete_user_query=db.delete("userdetails", "rollno=?", new
```

```
                String[]{String.valueOf(rollno)});
            if(delete_user_query >=0) {
                return true;
            }
            else{
                return false;
            }
        }
        else{

            return false;
        }
    }
}
```